# Beta Testing Document

**for**

# PlaceIIT

**Version 1.0**

**Prepared by Team Kernel Panic**


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**Course:** CS253

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# Revisions

| Version | Primary Author(s) | Description of Version | Date Completed |
|---|---|---|---|
| Version 1.0 | Abhinav Chhabra<br>Aman Srivastava<br>Dhruvi Jain<br>Garvit Bhutani<br>Hemakshi Kumawat<br>Laksh Dhir<br>Madhusudan Kantharia<br>Ravipati Raj Sri Vardhan<br>Shreyas Asthana<br>Sujal Satish Motagi | | 17/04/2026 |

# 1 Introduction

**PlaceIIT** is a browser-based application designed to manage and coordinate the dynamic activities that take place within the campus during internship and placement seasons under the Student's Placement Office (SPO).

The system is built to capture and reflect updates the moment administrators or coordinators receive new information, ensuring that all stakeholders remain synchronized without delays or confusion. It provides students with continuous visibility into interview queues, schedules, and status changes, reducing the need for repeated physical checks across venues.

Coordinators are equipped to manage interview rounds, candidate movement, and company-specific workflows through structured dashboards, while administrators maintain centralized control over overall placement operations. The platform also supports structured data ingestion, allowing company details, student shortlists, and logistical requirements to be uploaded and processed efficiently, thereby reducing manual overhead and minimizing the chances of inconsistency.

Overall, **PlaceIIT** serves as a unified digital layer that brings clarity, responsiveness, and operational discipline to the often fast-moving landscape of campus placements.

# 2 List of Reported Bugs

**BUG 1:**

**Tested Feature:** User Management / Administrative Role Assignment (APC Addition)

**Tester Name:** Shreyas Asthana

**Testing Date:** April 9, 2026

**Bug Details:** The system fails to validate the "Name" field when adding a new Assistant Placement Coordinator (APC). Specifically, the input validation logic does not check for empty strings or whitespace-only entries.

- **Test Case 1:** Navigate to the APC addition module. In the name field, enter " " (a single space). Observe that the system accepts the input and successfully creates an APC entry with a blank name.
- **Test Case 2:** Leave the APC name field entirely empty "". Observe that the system treats this as a valid entry rather than triggering a "Field Required" validation error.

**Bug Report Date:** April 9, 2026

**Has the bug been fixed?** Yes

**Date of Bug fixing:** April 12, 2026

**Additional comments:** The issue was addressed promptly by the development team (srv-05) within three days of reporting. The fix ensures data integrity within the administrative dashboard, preventing "ghost" accounts or null references in the coordinator list. The quick turnaround is satisfactory, though more robust client-side validation should be implemented across all "Addition" forms (like CoCo and Company entries) to prevent similar regressions.

**BUG 2:**

**Tested Feature:** Administrative APC Addition / Input Validation Logic

**Tester Name:** Shreyas Asthana

**Testing Date:** April 9, 2026

**Bug Details:** The manual entry form for adding a new Assistant Placement Coordinator (APC) lacked strict data validation and type-checking for the "Institute ID" and "Email" fields. The system accepted arbitrary strings that did not conform to the expected institutional or standard formats.

- **Test Case 1 (Email Validation):** Entering a nonsense string with repetitive extensions (e.g., a@b.comcc.comc.comcocmcomcom) was accepted by the system, indicating a lack of Regex-based email validation.
- **Test Case 2 (Institute ID Validation):** The "Institute ID" field accepted email-formatted strings (e.g., uu@uu.com) or any random alphanumeric string, rather than enforcing the numeric/specific format used by IITK.

**Bug Report Date:** April 9, 2026

**Has the bug been fixed?** Yes

**Date of Bug fixing:** April 12, 2026

**Additional comments:** The developer (noah-9) successfully implemented domain-level restrictions, limiting accepted emails to @iitk.ac.in or @gmail.com. This is a critical fix for PlaceIIT, as it prevents the creation of administrative accounts using unverified or external domains thereby enhancing the security of the RBAC (Role-Based Access Control) system. However, to fully close this loophole, we recommend adding character-length limits to the Institute ID field and a frontend verification to ensure the ID matches the expected student/staff format to prevent typos during manual ingestion.

**BUG 3:**

**Tested Feature:** User Profile Management / Profile Editing

**Tester Name:** Shreyas Asthana

**Testing Date:** April 9, 2026

**Bug Details:** The "Edit Profile" functionality lacked essential input sanitization and validation for the mobile number field. Users were able to bypass numerical constraints and save invalid data to their profiles.

- **Test Case 1 (Type Validation):** Changing the mobile number to an alphabetic string (e.g., "abcde") was permitted by the system without throwing a type-mismatch error.
- **Test Case 2 (Length/Format Validation):** Entering random sequences of numbers that do not correspond to standard 10-digit Indian mobile formats (e.g., "123" or "123456789012345") were accepted and updated in the database.

**Bug Report Date:** April 9, 2026

**Has the bug been fixed?** Yes

**Date of Bug fixing:** April 12, 2026

**Additional comments:** This was a high-priority issue as mobile numbers are likely used for coordination or potential OTP services within PlaceIIT. The fix by srv-05 ensures that the user profile maintains data integrity. For future iterations, we suggest implementing a regex that specifically enforces the +91 prefix to prevent accidental typos that could lead to a communication gap between students and Placement Coordinators during live interview rounds.

**BUG 4:**

**Tested Feature:** Company Coordinator (CoCo) Profile / Contact Functionality

**Tester Name:** Madhusudan Kantharia

**Testing Date:** April 9, 2026

**Bug Details:** The "Contact Admin" button located within individual Company Coordinator (CoCo) profiles in the CoCo management section (APC portal) was non-functional.

- **Test Case 1:** Upon navigating to a CoCo profile and clicking the "Contact Admin" button, the system failed to trigger any action (e.g., opening a mailto link, redirecting to a contact form, or launching an internal messaging modal). The button was a dead UI element effectively.

**Bug Report Date:** April 9, 2026

**Has the bug been fixed?** Yes (via feature removal)

**Date of Bug fixing:** April 12, 2026

**Additional comments:** The developer (srv-05) chose to resolve this by removing the button entirely rather than implementing the backend logic. While this removes the "dead link" from the UI and improves the immediate user experience by eliminating frustration, it leaves a potential functional gap since the feature is not available to CoCos. If direct communication between CoCos and Admins is a requirement for real-time coordination, such as reporting urgent interview venue changes, a working alternative (like a centralized "Help" or "Escalation" tab) in the CoCo portal should be prioritized in the next sprint to replace the removed functionality.

**BUG 5:**

**Tested Feature:** Company Management / "Add Company" Form

**Tester Name:** Madhusudan Kantharia

**Testing Date:** April 9, 2026

**Bug Details:** The input fields for "Company Name" and "Venue" lacked character encoding restrictions, allowing the submission of non-alphanumeric characters, specifically emojis.

- **Test Case 1:** During the "Add Company" process, entering emojis into the name and venue fields was permitted by the system. This results in unprofessional data representation on the student dashboard and potential rendering issues across different browser environments or mobile devices.

**Bug Report Date:** April 9, 2026

**Has the bug been fixed?** Yes

**Date of Bug fixing:** April 12, 2026

**Additional comments:** The developer (Schrodinger-Capy) implemented a fix to block emoji input, which is essential for maintaining the professional integrity of the PlaceIIT platform. However, the validation logic remains too permissive. For instance, the system currently accepts strings like ".." or "---" as valid company names. It would be highly beneficial to implement stricter string checking or a verification layer to ensure a legitimate company exists. Since this platform serves as a centralized coordination layer for IITK placements, preventing "junk" data entry is critical to avoid misleading students who rely on real-time data synchronization for their interview schedules.

**BUG 6:**

**Tested Feature:** User Security / Password Management (APC Password Update)

**Tester Name:** Madhusudan Kantharia

**Testing Date:** April 9, 2026

**Bug Details:** The password update mechanism for Assistant Placement Coordinators (APCs) lacked standard character set restrictions. Specifically, the system permitted the use of emojis within the password field.

- **Test Case 1:** When changing or setting an APC password, the tester successfully input and saved a string consisting entirely of emojis. While emojis are technically valid Unicode characters, allowing them in passwords without explicit intent can lead to cross-platform compatibility issues, authentication failures on devices with different emoji rendering engines, and potential security bypasses if sanitization is inconsistent.

**Bug Report Date:** April 9, 2026

**Has the bug been fixed?** Yes

**Date of Bug fixing:** April 12, 2026

**Additional comments:** The developer (Schrodinger-Capy) has now restricted emoji input for passwords. This is a vital step in maintaining a robust security posture for the PlaceIIT platform, especially given that APCs hold high-level administrative privileges. To further enhance security, the system should move beyond just blocking emojis and implement a comprehensive password policy—requiring a mix of uppercase letters, lowercase letters, numbers, and standard special characters—to ensure that all administrative accounts are protected against simple brute-force or dictionary attacks.

**BUG 7:**

**Tested Feature:** Company Coordinator (CoCo) Management / CoCo Addition

**Tester Name:** Abhinav Chhabra

**Testing Date:** April 9, 2026

**Bug Details:** The "Roll Number" input field in the CoCo addition form lacked strict numerical or format-based validation. The system allowed the entry of non-relevant data types without returning an error.

- **Test Case 1:** Inputting a purely alphabetic string (e.g., "NotARollNumber") into the Roll Number field. The system accepted the input as valid.
- **Test Case 2:** Inputting an invalid or out-of-range numerical sequence. The system failed to verify the entry against the standard IITK roll number format, leading to potential database inconsistencies.

**Bug Report Date:** April 9, 2026

**Has the bug been fixed?** Yes (Partial)

**Date of Bug fixing:** April 12, 2026

**Additional comments:** While the developers (noah-9 and Schrodinger-Capy) have implemented constraints to ensure the field only accepts numeric input and rejects random alphabetic strings, the validation logic remains incomplete. The field currently accepts numerical entries that exceed the maximum legitimate digits possible for an IITK roll number. Since roll numbers are critical for mapping students to interview rounds and real-time status visibility, it is vital to implement a strict length constraint (e.g., exactly 6 or 8 digits) and range checking. This would prevent the creation of non-existent coordinator profiles that could disrupt the centralized data management system.

**BUG 8:**

**Tested Feature:** Company Coordinator (CoCo) Management / CoCo Addition

**Tester Name:** Abhinav Chhabra

**Testing Date:** April 9, 2026

**Bug Details:** The "Name" field in the Company Coordinator (CoCo) addition module lacked mandatory string validation. The system allowed users to be created with non-identifiable name strings, which bypasses the purpose of role-based dashboards and record-keeping.

- **Test Case 1:** In the name input field, the tester entered " " (a single space). The system accepted this as a valid entry and successfully added a CoCo with a blank name.
- **Test Case 2:** Leaving the field entirely empty (null string) also allowed the submission to proceed, failing to trigger a "Required Field" alert.

**Bug Report Date:** April 9, 2026

**Has the bug been fixed?** Yes

**Date of Bug fixing:** April 12, 2026

**Additional comments:** The developer (Schrodinger-Capy) successfully implemented whitespace trimming and mandatory string checks to prevent empty entries. However, since PlaceIIT is designed for centralized coordination within IITK, a higher-level recommendation would be to implement a cross-reference check against a master student database. Currently, an admin could manually enter any alphabetical string. Linking the "Add CoCo" feature to the existing student database via Roll Number or LDAP would automate name retrieval, ensuring that only verified students are assigned as coordinators. While this increases database management complexity, it is necessary to eliminate the information gap and ensure 100% data accuracy for real-time interview status visibility.

**BUG 9:**

**Tested Feature:** User Security / Contact Information Update (Change Password & Contact Details)

**Tester Name:** Dhruvi Jain

**Testing Date:** April 9, 2026

**Bug Details:** The profile update form, which requires users to complete contact details during a password change, lacked validation for the "Emergency Contact Name" and "Friend Contact Name" fields. This allowed the submission of non-alphabetic, random strings.

- **Test Case 1:** Inputting a sequence of special characters and symbols (e.g., ---- T_T -------) into the Emergency Contact Name field.
- **Test Case 2:** Inputting an arbitrary string (e.g., ((((((( ++ )))))))) into the Friend Contact Name field. In both cases, the system accepted these inputs as valid names, which renders the emergency contact feature useless for actual administrative or safety coordination.

**Bug Report Date:** April 9, 2026

**Has the bug been fixed?** Yes

**Date of Bug fixing:** April 12, 2026

**Additional comments:** The developer (Schrodinger-Capy) successfully restricted these fields to only accept letters and spaces. This is a critical fix for the safety requirements of the platform, as emergency contacts must be identifiable individuals. However, we noticed that in the failing test case, the phone number fields were also populated with placeholder/invalid data (e.g., 0909090909). We also recommend extending the validation logic to ensure these phone numbers are exactly 10 digits and do not consist of repetitive or obviously dummy sequences. Incorrect emergency data could lead to serious communication failures during placement-related emergencies.

**BUG 10:**

**Tested Feature:** Add Student (Form Validation)

**Tester Name:** Dhruvi Jain

**Testing Date:** April 9, 2026

**Bug Details:** The system accepts invalid entries when admin manually adds students. Specifically, emojis are allowed in the "Name" field, and the "Email ID" field accepts any string without enforcing standard email format restrictions.

- **Test Case 1:** Navigate to the form to manually add a student. Enter emojis in the "Name" field. Observe that the form submits successfully without validation errors.
- **Test Case 2:** Enter an arbitrary string without an '@' symbol in the "Email ID" field. Observe that the form submits successfully without validation errors.

**Bug Report Date:** April 9, 2026

**Has the bug been fixed?** Yes

**Date of Bug fixing:** April 12, 2026

**Additional comments:** Proper form validation was added to restrict invalid characters (like emojis) in the “Name” field. Correct email string formatting. However, the “Roll no.” field still accepts strings apart from numbers, and needs to be fixed.

**BUG 11:**

**Tested Feature:** Manual Shortlisting of Students (UI Overlay)

**Tester Name:** Dhruvi Jain

**Testing Date:** April 9, 2026

**Bug Details:** While adding shortlisted students, if admin selects multiple students from the search suggestions, the dropdown overlay blocks the UI and cannot be exited or dismissed, making it impossible to click the "Add Students" button.

- **Test Case 1:** Open the "Add Students Manually" modal. Search for students and select multiple names from the dropdown menu. Observe that the window cannot be exited (even by pressing enter) to click the add button.

**Bug Report Date:** April 9, 2026

**Has the bug been fixed?** Yes

**Date of Bug fixing:** April 12, 2026

**Additional comments:** A "Close Suggestions" option was added to the UI, allowing users to dismiss the dropdown overlay after selecting multiple students. We recommend implementing a "click outside" event listener to automatically dismiss the overlay when the user clicks anywhere else on the screen, and allow users to press the "Esc" key to close the dropdown for improved accessibility and smoother UX.

**BUG 12:**

**Tested Feature:** APC Profile Management

**Tester Name:** Hemakshi Kumawat

**Testing Date:** April 9, 2026

**Bug Details:** The profile fields for an Assistant Placement Coordinator (APC) are completely uneditable, despite being designed to allow edits.

**Test Case 1:** Log in as an APC. Navigate to "My Profile". Attempt to edit the APC profile details. Observe that nothing can be edited.

**Bug Report Date:** April 9, 2026

**Has the bug been fixed?** Yes

**Date of Bug fixing:** April 12, 2026

**Additional comments:** Fixed. The developer clarified and corrected the permissions so that only the password and phone number fields are editable for an APC profile.

**BUG 13:**

**Tested Feature:** Company Venue Management (Edit Venue Input)

**Tester Name:** Hemakshi Kumawat

**Testing Date:** April 9, 2026

**Bug Details:** The system accepts empty strings or single spaces as valid inputs for the venue location while editing the venue of a company.

- **Test Case 1:** Navigate to the company overview card. Click to edit the venue of the company. Leave the venue field as a blank space " " and save. Observe that the system accepts the empty string/space.

**Bug Report Date:** April 9, 2026

**Has the bug been fixed?** Yes

**Date of Bug fixing:** April 12, 2026

**Additional comments:** Input validation was implemented to prevent the system from accepting empty strings or spaces as a valid venue. We recommend considering changing the venue input from a free-text field to a standardized dropdown list of valid, predefined campus locations to eliminate typos entirely. Alternatively, enforce strict alphanumeric validation and character limits.

**BUG 14:**

**Tested Feature:** APC Notification System (Shortlisting & CoCo Assignment)

**Tester Name:** Hemakshi Kumawat

**Testing Date:** April 9, 2026

**Bug Details:** No real-time or system notifications are received by the assigned CoCo or the shortlisted student to inform them of the update when an APC shortlists a student or assigns a CoCo to a company.

- **Test Case 1:** Log in to the APC portal and shortlist a student to a company. Check the respective student's account for alerts. Observe that no notification is received.
- **Test Case 2:** Manually assign a CoCo to a company. Check the respective CoCo's account for alerts. Observe that no notification is received.

**Bug Report Date:** April 9, 2026

**Has the bug been fixed?** Yes

**Date of Bug fixing:** April 12, 2026

**Additional comments:** Fixed. The system now correctly triggers and sends notifications to the respective individuals upon assignment or shortlisting. We recommend enhancing these notifications by including action buttons or deep links within the alert itself. This would allow the recipient to navigate directly to the relevant company dashboard or view their new assignment details with a single click.

**BUG 15:**

**Tested Feature:** Login Authentication & Role Switching

**Tester Name:** Aman Srivastava

**Testing Date:** April 9, 2026

**Bug Details:** When a user enters invalid credentials on one of the login pages (e.g., the student login page), and subsequently opens another login page (CoCo or APC/Admin login), clicking the back button to return to the first page causes the invalid credentials flag to still appear, even without entering any credentials.

- **Test Case 1:** Navigate to the Student Login page. Enter invalid credentials and submit the form. Observe the "Invalid credentials" error message. Navigate to a different login page. Click the browser's back button to return to the Student Login page. Observe that the "Invalid credentials" flag persists on the screen.

**Bug Report Date:** April 9, 2026

**Has the bug been fixed?** Yes

**Date of Bug fixing:** April 12, 2026

**Additional comments:** The state management was updated to ensure the invalid credentials flag is cleared when switching roles or navigating back between login views.

# 3 Overall Quality of the Software

**1.How good is the user manual? Was it easy to install and run the software?**
The user manual is quite good overall. It is thorough, role-based, and clearly organized into sections for students, CoCos, and APCs, making it useful for different types of users. It also does a solid job of defining system terms and explaining workflows step by step, so someone new to the platform would have enough guidance to understand how the system is meant to be used. At the same time, it is very detailed and sometimes a bit dense, so it feels more like a full reference guide than a quick-start manual.

It was indeed easy for us to install the software and work on it. It worked without any friction and ran very smoothly. The manual says PlaceIIT is web-based and does not require any standalone desktop software, so users only need a supported browser and the provided URL to access it. It also notes that the platform works on common operating systems and that file uploads happen through the browser without extra plugins, which makes setup straightforward.

**2.How is the quality of the code? Is it well documented? Is the code modular? Are the variables and the function names meaningful?**

The overall quality of the code appears to be good. It is reasonably well structured, with the frontend and backend separated clearly and different features organized into dedicated folders, which shows a modular design approach. This makes the project easier to maintain, understand, and extend over time. The documentation is also decent, with setup instructions and project details that help new users or developers get started without much confusion.

Variable names and function names seem meaningful in most places, following practical naming conventions that reflect their purpose. This improves readability and makes the code easier to follow. While there may be some areas that could be refined further to maintain consistency, overall, the codebase gives the impression of being thoughtfully written and organized to a good standard.

**3.Do you think that the software lacks some major features that may make it less acceptable to the use**

The software is well-designed and covers most of the essential features required for efficient placement coordination, including real-time queue management and role-based access. It provides a strong foundation for handling placement activities in a structured manner. However, it still lacks some important features that could further enhance its usability and effectiveness, which have been discussed below.

- **Lack of Recruiter / Company Interface:** The system currently supports only students, coordinators, and admins, but does not include a dedicated interface for

recruiters. This limits direct interaction between companies and the platform, forcing reliance on coordinators for communication and updates. A recruiter dashboard could allow companies to view candidates, provide feedback, and update results in real time. Without this, the system remains internally efficient but not fully integrated with all stakeholders.

- **Absence of Emergency Handling / Exception Management:** The system assumes a smooth flow of interviews but does not account for real-world exceptions like delays, overlapping interviews, or student unavailability. There is no provision for students to request rescheduling or for coordinators to dynamically adjust priorities in urgent cases. This can lead to confusion and missed opportunities despite having a structured queue system. Adding exception handling would make the system more robust and realistic.

- **Lack of Feedback and Review System:** The system does not include any mechanism for collecting feedback from students, coordinators, or recruiters after interviews or placement processes. This limits the ability to evaluate user experience, identify issues, and improve the system over time. A feedback system could help gather insights on interview processes, coordinator efficiency, and platform usability. Continuous improvement and quality assessment of the system remain difficult without it.

**4.Does the software not satisfy any major non-functional requirements?**

The software appears to have a weakness in session management. Even after several days of inactivity, the user session remained active and the website was still logged in automatically. This suggests that authentication tokens or sessions may not expire appropriately after a reasonable period. From a non-functional perspective, this does not fully satisfy security requirements, as placement-related systems should implement proper session timeout, token expiry, and automatic logout after inactivity to reduce the risk of unauthorized access on shared or unattended devices. Other than this issue, the software generally seems to satisfy most major non-functional requirements reasonably well. The platform appears usable, responsive, and accessible through a web browser without complicated installation. Overall, the system performs well, with session security being the main noticeable area for improvement.

# Appendix A - Group Log

Following the finalization of the project scope and requirements analysis, the team maintained continuous coordination through dedicated communication channels (WhatsApp and in-person meetings) to refine the system implementation, testing strategy, and documentation. The details of these discussions and collaborative efforts are summarized in the table below.

| Meeting Date and Time | Discussions and Outcomes |
|---|---|
| 9th April 2026 (14:00 - 16:30) | Commenced comprehensive system testing and bug identification across all major modules. Team members assigned testing responsibilities for user management, company coordination, and profile editing features. Documented initial findings and created preliminary bug report templates for standardized reporting. |
| 10th April 2026 (10:00 - 12:00 & 16:30 - 18:00) | Continued systematic testing of input validation mechanisms and security features. Reviewed individual bug reports submitted by team members and discussed severity levels. Coordinated with the development team (srv-05, noah-9, Schrodinger-Capy) to prioritize critical issues affecting user authentication and data integrity. |
| 11th April 2026 (13:00 - 15:30 & 17:00 - 19:00) | Conducted verification testing on fixes implemented by the development team. Analyzed the effectiveness of applied patches and identified any edge cases or regression issues. Prepared comprehensive documentation of all identified bugs with detailed test cases, reproduction steps, and impact assessments for final report compilation. |
| 12th April 2026 (11:00 - 13:00 & 15:00 - 17:00) | Finalized the beta testing document and consolidated all findings into the formal report. Conducted a comprehensive review of the overall software quality, assessed the completeness of implemented features, and evaluated the platform's readiness for production deployment. Team members provided recommendations for future improvements and optimization. |

**Team Members Involved:**

Abhinav Chhabra (240031), Aman Srivastava (240103), Dhruvi Jain (240359), Garvit Bhutani (240393), Hemakshi Kumawat (240447), Laksh Dhir (240588), Madhusudan Kantharia (240606), Ravipati Raj Sri Vardhan (240857), Shreyas Asthana (241001), Sujal Satish Motagi (241052)